

Effects of Line Length, Line Spacing, and Line Number on Proofreading Performance and Scrolling of Chinese Text

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Objective: The main purpose of this study was to investigate the effects and interactions of line length, line number, and line spacing on Chinese screen-based proofreading performance and amount of scrolling.

Background: Proofreading is an important process, and much of it is now done on screen. The Chinese language is increasingly important, but very little work has been done on the factors that affect proofreading performance for Chinese passages.

Method: Three display factors related to screen size, namely line length, line number, and line spacing, were selected to be investigated in an experiment to determine their effects on proofreading performance and amount of scrolling. Correlations between proofreading performance in time and accuracy and scrolling amount were also analyzed.

Results: The results showed that line number and line spacing had significant main and interaction effects on both proofreading time and detection rate. Line length and line number influenced scrolling amount significantly, but there was no interaction effect for scrolling. Scrolling amount was negatively correlated with proofreading time and typo detection rate such that more scrolling movement was associated with faster proofreading, but lower detection rate. There was a trade-off between time and accuracy.

Conclusion: For balancing time and detection rate and improving performance for on-screen Chinese proofreading, the display setting of medium line length (36 characters per line) with four lines and 1.5 line spacing should be used.

Application: The findings provide information and recommendations for display factors and the screen design that should prove useful for improving proofreading time and accuracy.

Keywords: proofreading, Chinese, screen design, scrolling

INTRODUCTION

The population of China is around 1.3 billion, and the Chinese language is used extensively in different kinds of publications (Rosenberg, 2010). It is very likely that the number of people using the Chinese writing system will increase rapidly due to the large Chinese population and the spread of Chinese influence. An unabridged Chinese dictionary contains about 48,000 Chinese characters (*Zhonghua da zidian*, 1978) and 2,500 of them are most frequently used (Ministry of Education of the People's Republic of China, 2005). Unlike English, a very large number of individual characters can act as words and can also be combined to form words and sentences. Each Chinese word is formed by one to five characters in general, and the frequency of usage of one- and two-character Chinese words in texts is about 98.6% (Lua, 1990).

Information technology has developed rapidly, and the accessibility of the Internet has changed people's reading habits. Due to the flexibility and convenience of electronic documentation, the employment of computer screens for reading has increased significantly, and Internet users show a preference for reading many types of written material on computer screens (Shaikh & Chaparro, 2004). For example, e-books are gaining in popularity, thereby increasing the need for proofreading. Project Gutenberg was initiated to digitize, archive, and distribute mainly full texts of public domain books along with a web-based scheme to proofread drafts of e-texts for errors. Compared with the 30,768 English releases, the 405 finished Chinese items is much fewer (Project Gutenberg News, 2012). If the display format of the drafts and the screen design for Chinese can be made more user-friendly, proofreading would be improved and a larger number and variety of items could become available to a larger audi-

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ence. Performance in Chinese communication tasks has been investigated quite recently (Chan & Lee, 2005a, 2005b, 2005c; Chan & So, 2009; Lee & Chan, 2003; Li, Zhang, Zhou, Yang, & Li, 2012; So & Chan, in press; Zhao, Rau, & Yang, 2005), but not a great deal has been done to study proofreading.

Proofreading is the detection of errors in text and involves careful reading of a passage and marking deviations from specifications, such as typeface, misspellings, nonstandard grammar, and many other errors in language on a newly produced version, called the live copy (Smith, 1993). Comparison and noncomparison are the two main methods of proofreading (Anderson, 1990). For noncomparison proofreading there is usually no true dead copy (original version) or there is a need to refer to the dead copy only when the proofreader sees something puzzling. The proofreader checks for errors and essentially ensures that the document is readable and correct. For comparison proofreading, the dead copy and live copy are directly compared to each other. Comparison proofreading is a critical process for creating an electronic book from a physical book and is done after optical character recognition on the physical book is performed (Anderson, 1990). Chinese characters are much more complex structures than are Latin scripts, so repeated proofreading is needed after scanning or manual input (Yang, 2007).

The display factors of font size and typeface for English (Bernard, Chaparro, Mills, & Halcomb, 2003) and Korean characters (Kong, Lee, Jung, & Song, 2011) have been shown to affect text readability and reading task efficiency on computer screens. Research studies on Chinese characters with regard to legibility assessment (Chi, Cai, & You, 2003; Lin, Lin, Hwang, Jeng, & Liao, 2009; Wu, 2011), reading performance (Huang, Rau, & Liu, 2009; Lai, 2008), and character identification task (Shen, Shieh, Chao, & Lee, 2009; Yau, Chao, & Hwang, 2008) have been reported. Chan and Ng (2012) did a study for Chinese comparison proofreading and found that the typeface Ming Liu with 14-point font size produced faster proofreading speed than other combinations. For Chinese proofreading, unlike for English, text direction is a factor to be considered. Chinese text has been traditionally

written and printed vertically from the top to the bottom of a page, with columns moving from right to left. However, there has been a trend to print material horizontally (usually starting in the top left corner) since the early 20th century. Sun, Morita, and Stark (1985) found that reading horizontal Chinese characters was twice as fast as reading vertical ones. In a study of Chinese menu orientation, Shih and Goonetilleke (1998) found that a one-row word menu was more effective than a one-column word menu in terms of search time and accuracy. It was therefore expected that the horizontal reading direction would be more preferred and could result in better Chinese proofreading performance than the vertical direction. This was confirmed by Chan and Ng (2012) in a recent proofreading experiment that the horizontal text direction produced faster proofreading and better subjective experience than the vertical ones. The benefit of top-bottom copy placement setting for use in Chinese proofreading interfaces for increasing accuracy was also confirmed in that study.

Other than the previously studied factors, the amount of textual information presented and display settings related to the size and layout of the screen must also be considered. There are many display factors involved in textual presentation, which, for example, include line length, line number, and line spacing (Elkerton & Williges, 1984; Hoffman, Lee, & McGehee, 2006; Ling & van Schaik, 2006, 2007). The paged view and continuous scrolling view are two common ways of displaying a large amount of text on screen (Baker, 2003; Braganza, Marriott, Moulder, Wybrow, & Dwyer, 2009; Hsieh, Lin, Chen, Huang, & Chen, 2007; Sanchez & Wiley, 2009). For reading style, Zhang, Zhang, Bin, Zhang, and Zhou (2008) found that people completed Chinese information search tasks more quickly using the scrolling-by-line style and comprehended a reading task much better with the scrolling-by-page style. However, a recent study revealed that the execution times for within document searching tasks under scrolling and pagination mode were similar to each other (Marcial & Hemminger, 2011).

Previous research examining the number of lines of text in a paged view has produced varied results in reading tasks (Dyson, 2004). For

presenting a large unit of text, the greater the numbers of lines shown within a window, the lower the frequency of scrolling needed and thus scrolling time is shorter. However, the crowding effect is a problem that will occur if a large amount of text is confined and cramped in a limited space. Ling and van Schaik (2007) found that wider line spacing resulted in better accuracy and faster search times. Previous work on line length revealed that there are two opposing factors for determining the optimum line length for English reading. If line length is too short, not much information can be extracted from the effective visual field during each eye fixation and reading is particularly inefficient. In general, longer lines (between 75 and 100 English characters per line) were read faster than very short lines (Dyson & Kipping, 1998). However, if the lines are too long, readers may take more time to locate the beginning of the following line with the return eye sweep. Dyson and Haselgrove (2001) later found that a line length of 55 characters produced the highest level of comprehension and provided conditions for effective reading at normal and fast speeds. Ling and van Schaik (2006) suggested that longer line length (85–100 characters) should be used for quick scanning of information, whereas shorter line lengths (55–70 characters) were for text requiring precise reading. In a recent study of the effects of typographical factors on reading and comprehension performance in Arabic, Ganayim and Ibrahim (2012) found that participants' comprehension scores were better with the long line length setting. However, the optimum line length for reading or proofreading Chinese is not yet known.

In addition to the amount of textual information on the screen, the amount and characteristics of scrolling will be affected by the size and layout of the screen (De Bruijn, de Mul, & van Oostendorp, 1992). A larger screen size may require less scrolling time and reduce text manipulation (Dyson & Kipping, 1998), but the greater amount of text may increase eye movement time between live and dead copies. On a smaller screen, the need for more scrolling movement may increase the difficulty of comparing and locating the text (Marcial & Hemminger, 2011; O'Hara & Snellen, 1997). Such

considerations indicate that there might be interrelationships among the display factor settings, proofreading performance, and amount of scrolling required.

Proofreading on computer screens is increasingly common because of the convenience of manipulating documents on computers. Therefore, it is important to determine optimum design and layout of screens to facilitate Chinese proofreading. In the research presented here, three display factors related to screen size, namely line length, line number, and line spacing, were selected to be investigated to determine their effects on proofreading performance and amount of scrolling. Line number referred to the number of visible lines in the window view at a time in this study. It was expected that proofreading task performance and text manipulation characteristics would vary with these screen design parameters. The larger the number of lines and the smaller the line spacing, the greater the expected effect of text crowding. Considering the need for the eyes to locate the beginning of the next line in return sweeps, there should be an ideal for the displayed line length in the window texts. All these considerations provided the context and basis for this study, and it was hoped that some optimum screen design parameters for Chinese proofreading could be recommended.

METHOD

Design

Previous studies of proofreading showed that the typeface Ming Liu with 14-point font size together with text presented horizontally with an above-below copy placement arrangement produced better performance in proofreading Chinese passages (Chan & Ng, 2012). The better performance with the above-below copy placement than left-right placement was shown to be true even with the shorter line widths in the side-by-side arrangement. Three levels of each selected display factor were tested. For line spacing, single (1), one and a half (1.5), and double (2) spacing were used. For line number, two, four, and eight lines were tested (Figure 1). The above-below copy placement, the 14-point font size, and the maximum two-line spacing physically limited the maximum

咱們畫圖，有時候為的實用。編撰關於動物植物的書籍，要讓讀者明白動物植物外面的形態跟內部的構造，就得畫種種

2 Lines

咱們畫圖，有時候為的實用。編撰關於動物植物的書籍，要讓讀者明白動物植物外面的形態跟內部的構造，就得畫種種動物植物的圖。修建一所房屋或者佈置一個花園，要讓住在別地的朋友知道房屋花園是怎麼個光景，就得畫關於這所房

4 Lines

咱們畫圖，有時候為的實用。編撰關於動物植物的書籍，要讓讀者明白動物植物外面的形態跟內部的構造，就得畫種種動物植物的圖。修建一所房屋或者佈置一個花園，要讓住在別地的朋友知道房屋花園是怎麼個光景，就得畫關於這所房屋這個花園的圖。這類的圖，繪畫的動機都在實用。讀者看了，明白了；住在別地的朋友看了，知道了，就完成了它的功能了。這類圖不能隨便亂畫。首先把畫的東西看明白，認得確切。譬如畫貓罷，牠的耳朵怎麼樣，牠的眼睛怎麼樣，

8 Lines

Figure 1. Textual presentation with different line numbers.

number of lines to be shown. It was known that crowding and increased text density would lead to decreased performance; with the current display parameters, a maximum of eight lines was thus determined to be tested. For line length, the three levels were short (S), medium (M), and long (L), with 26, 36, and 46 characters per line, respectively (Figure 2). The three display factors were studied with a factorial design. There were altogether 27 (3 line lengths $\times$ 3 line numbers $\times$ 3 line spacings) different display conditions. To avoid the effect of both physical and mental fatigue on experimental results, line length was selected as a between-participants factor. Therefore, 39 participants were divided into three equal groups of 13 and assigned to the three levels of line length. As a result, each participant was tested with 9 (3 line numbers $\times$ 3 line spacings) display conditions. To ensure the reliability of data, each participant was tested with the nine display conditions twice, with the sequence reversed on two consecutive days. Altogether 18 different passages were presented for the nine conditions for each participant. The 18 passages were totally different in content, eliminating the memory problem. Nine passages were proofread in a day, and a 2-min break was given for reading every three passages

for avoidance of mental or visual fatigue. For participants to proofread the passages, three of the most commonly available scrolling methods were used, namely, (a) the rotating scrolling wheel on a computer mouse, (b) manipulating an on-screen scroll bar, and (c) pressing the up and down buttons on a keyboard. The amount, patterns, and preferences of scrolling for each of the three methods were recorded and analyzed. Figure 3 shows the general layout of the display. Due to the need of concurrent presentation of the live and dead copies on screen in ordinary proofreading environment (Newby & Franks, 2003), smaller window sizes were used in the proofreading interfaces compared to those used in the reading tasks.

Participants

In all, 39 native Chinese-speaking undergraduates (27 males and 12 females) participated in the experiment voluntarily. The age range was 21 to 26 years old ($Mdn = 22$). Participants had computer-using experience ranging from 8 to 15 years ($M = 11$ years). They were all self-reported right-handers and used their right hands for operating a computer mouse and for handwriting.

Materials

A total of 24 unrelated Chinese passages were randomly selected from the sample passages available for reference by the fifth-level secondary school students preparing for the Chinese Language Examination of the Hong Kong Certificate of Education Examinations. A focus group of five participants of the same education background as those tested in this study was asked to preview all the passages and select 18 passages of similar difficulty. The 18 passages with the highest frequencies of choices were used for testing. Including punctuation marks, the word counts of the 18 passages were between 844 and 934 ($M = 901$, $SD = 25$). Table 1 shows the number of lines for a passage for the different display conditions. All passages were used as dead copies (original version) and were modified to be used as live copies (newly typed versions) by embedding typographical errors (typos) in them. The seven commonly found typos used were an extra word, missing

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26 Characters per line

咱們畫圖，有時候為的實用。編撰關於動物植物的書籍，要讓讀者明白動物植物外面的形態跟內部的構造，就得畫種種動物植物的圖。修建一所房屋或者佈置一個花

36 Characters per line

咱們畫圖，有時候為的實用。編撰關於動物植物的書籍，要讓讀者明白動物植物外面的形態跟內部的構造，就得畫種種動物植物的圖。修建一所房屋或者佈置一個花園，要讓住在別地的朋友知道房屋花園是怎麼

46 Characters per line

Figure 2. Textual presentation with different line lengths.



Figure 3. Sample layouts of the text interfaces under the condition of long line length and four lines and 1.5 line spacing. Participants were asked to compare the dead copy with the live copy and to highlight each typo on the live copy with the corresponding button at the bottom of the screen. The buttons from left to right are for the indication of typos such as extra word, missing word, wrong word, wrong font type, wrong punctuation mark, wrong order, and extra spacing and to undo action. The rightmost one is for displaying the next passage.

word, wrong word, wrong font type, wrong punctuation mark, wrong order, and extra spacing. Each typo appeared 14 times randomly throughout the 18 passages, making a total of

98 typos. There were 3 to 8 typos in each passage. The number and types of typos were not known by participants, who were told that there was at least one typo in each passage.

TABLE 1: Number of Lines for a Passage Under Each Condition of Line Length and Line Spacing

Line Length	Line Spacing	Number of Lines for a Passage
Small	1	35
	1.5	52
	2	69
	2	69
Medium	1	25
	1.5	38
	2	50
	2	50
Long	1	20
	1.5	29
	2	39
	2	39

Apparatus

A personal computer and a 17-in. liquid crystal display monitor (1,280 × 1,024 pixels at 70 Hz refresh rate) were used to operate the Chinese Proofreading Program, which was prepared with Microsoft Visual Basic 2008. The monitor was placed on a table of 900 mm height and was 400 mm from participants’ eyes. At this viewing distance, the 14-point Ming Liu font size subtended a visual angle of 0.72° to participants’ eyes. A keyboard and a computer mouse with scrolling wheel were provided to participants for operating the proofreading task.

Procedure

Before the start of the experiment, the experimental procedure and types of typos were explained to participants with an instructional video. A 10-min practice session was given prior to the testing. Participants were asked to proofread the passages carefully and highlight the typos in the live copy as fast as they could and then click the corresponding typo type buttons at the bottom of the screen. When participants determined that all typos had been found, they could press the “Next Passage” button to proofread the next passage. A 2-min rest was given to participants after proofreading three passages to avoid mental or visual fatigue.

RESULTS

Proofreading Time

A total of 702 (39 participants × 9 conditions × 2 tasks) passage proofreading data points

TABLE 2: Proofreading Time (msC⁻¹) for Each Display Factor

Factor	Level	M	SD
Line length	S	304	99
	M	360	133
	L	340	122
Line number	2	320	107
	4	312	104
	8	372	140
Line spacing	1	315	110
	1.5	336	126
	2	352	124

were collected. The average proofreading time per character (msC⁻¹) for each display factor is shown in Table 2. The results of a three-way mixed analysis of variance (ANOVA) showed that there were significant line number, $F(2, 72) = 23.396, p < .001, \eta^2_p = .39$, and line spacing, $F(2, 72) = 15.565, p < .001, \eta^2_p = .18$, effects on proofreading time, whereas the line length effect was nonsignificant. For the line number factor, the shortest mean proofreading time (312 msC⁻¹) was for the four-line passages, whereas two-line passages had slightly longer times (320 msC⁻¹) and the eight-line passages resulted in the longest time (372 msC⁻¹). For line spacing, there was a trend showing proofreading time increased with increasing line spacing. Apart from the significant Line Number × Line Spacing interaction, $F(4, 144) = 2.725, p < .05, \eta^2_p = .07$, all of the remaining interactions of the three factors were nonsignificant. Figure 4 shows the significant Line Number × Line Spacing interaction plot.

Typo Detection Rate

The typo detection rate represented the percentage of correct detections per passage, and the average typo detection rate for each display factor is shown in Table 3. Similar to the result for proofreading time, a three-way mixed ANOVA showed that there were significant line number, $F(2, 72) = 21.784, p < .001, \eta^2_p = .38$, and line spacing, $F(2, 72) = 14.130, p < .001, \eta^2_p = .28$, effects on detection rate, but the effect of line length was nonsignificant. The pairwise comparisons for the main effect of line number

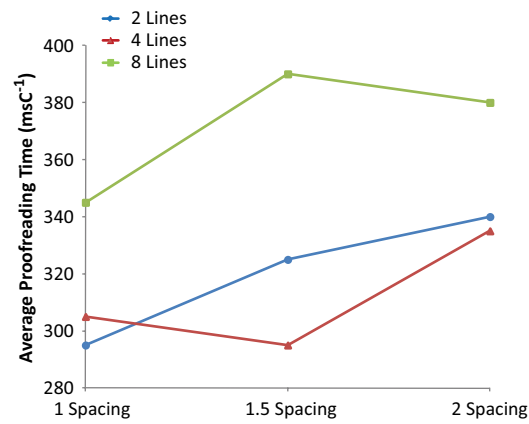


Figure 4. Average proofreading time as a function of line number and line spacing.

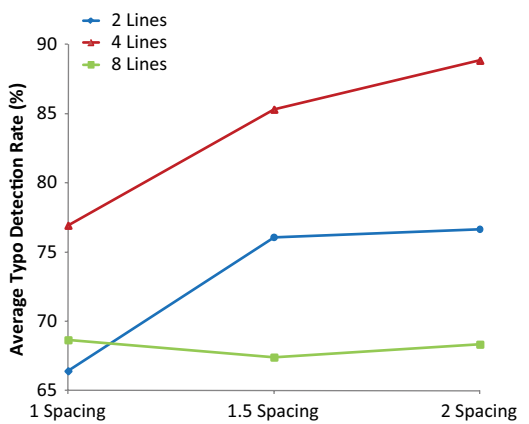


Figure 5. Average typo detection rate as a function of line number and line spacing.

TABLE 3: Typo Detection Rate (%) for Each Display Factor

Factor	Level	M	SD
Line length	S	72.80	22.10
	M	75.71	21.88
	L	76.34	19.85
Line number	2	73.04	20.26
	4	83.69	20.12
	8	68.13	20.64
Line spacing	1	70.66	21.81
	1.5	76.26	21.18
	2	77.94	20.37

corrected with Bonferroni adjustment showed significant difference in detection rates amongst the three line number conditions ($p < .05$, $\eta^2_p = .24$, $.53$, and $.76$, respectively). Average detection rate was highest (83.69%) for four-line passages, lowest (68.13%) for eight-line passages, and in between those values for two-line passages (73.04%). For the main effect of line spacing, the detection rates for 1.5 line spacing (76.26%) and double spacing (77.94%) were significantly higher than that for single spacing (70.66%; $p < .05$, $\eta^2_p = .26$ and $.34$, respectively). But there was no significant difference between 1.5 line spacing and double spacing.

For the interactions of the three display factors, again apart from the significant interaction

of line number and line spacing, $F(4, 144) = 4.812$, $p < .005$, $\eta^2_p = .12$, all of the remaining interactions were nonsignificant. Figure 5 shows the two-way interaction plot of average detection rate for the factors of line number and line spacing. For this interaction, the first contrast reflected a difference in detection rate between four-line and eight-line passages under single spacing and 1.5 line spacing ($p < .005$). In single spacing the detection rate for four-line passages (76.92%) was higher than that for eight-line passages (68.44%). Similarly, in 1.5 line spacing the detection rate for four-line passages (85.26%) was higher than that for eight-line passages (66.86%). It can be concluded that the increase in detection rate due to the use of four-line compared to eight-line passages was significantly greater when 1.5 line spacing instead of single spacing was used. The second contrast examined differences in typo detection rate between two-line and eight-line passages under single and double spacing ($p < .05$, $\eta^2_p = .21$) and between four-line and eight-line passages under single and double spacing ($p < .005$, $\eta^2_p = .31$). The detection rates for two-line (66.35%) and eight-line passages (68.44%) were similar under single spacing, whereas the detection rate for two-line passages (76.56%) was higher than eight-line passages (67.85%) in double spacing. The detection rate for four-line passages was higher than for eight-line passage under both single and double spacing.

TABLE 4: Amount of Scrolling (number of lines) for Each Display Factor

Factor	Level	M	SD
Line length	S	34.82	11.11
	M	25.83	8.29
	L	18.53	7.48
Line number	2	31.54	14.52
	4	25.45	7.46
	8	22.19	8.38
Line spacing	1	26.79	11.75
	1.5	26.02	11.54
	2	26.37	10.50

Number of Nondetected Typos and Related False Alarms

In addition to detection rate, the number of embedded typos that could not be detected (number of nondetected typos) and the number of places incorrectly identified as a typo (number of related false alarms) by participants were analyzed. The sum of the number of nondetected typos and number of related false alarms (NDTFA) was taken as a form of accuracy measure in this study. On average, the NDTFAs for a participant in the proofreading task for extra word, missing word, wrong word, wrong font type, wrong punctuation mark, wrong order, and extra spacing were 3.46, 3.13, 6.15, 0.82, 7.74, 5.67, 1.72, respectively.

Amount of Scrolling

The amount of scrolling represented the total number of lines moved by upward and downward scrolling movement per each passage with any of the three scrolling methods. Overall, 92.24% of scrolling movements were downward and only 7.76% were upward. Of the three scrolling methods, the participants had a strong preference for the scrolling wheel on the computer mouse (49.56%) and the screen scroll bar (41.70%), whereas the keyboard buttons were seldom used (8.74%). The average scrolling amount for each display factor is shown in Table 4. The results of three-way mixed ANOVA showed that there were significant line length, $F(2, 36) = 61.248, p < .001, \eta^2_p = .77$, and line number, $F(2, 72) = 56.347, p < .001, \eta^2_p = .61$,

effects on scrolling amount, whereas the effect of line spacing was nonsignificant. For line length, there was a steady decrease of scrolling amount from 34.82 lines to 18.53 lines with increasing line length. A similar trend was found for line number. Scrolling amount decreased from 31.54 lines to 22.19 lines with an increasing number of lines. There were no significant interaction effects for the three display factors for scrolling amount.

Correlations Among Proofreading Time, Typo Detection Rate, and Amount of Scrolling

Pearson’s correlations were calculated to evaluate the relationships among the main performance measures (Table 5). Proofreading time was found positively correlated with detection rate ($r = .171, R^2 = .029, N = 702, p < .001$). Scrolling amount was negatively associated with proofreading time ($r = -.109, R^2 = .012, N = 702, p < .01$) and detection rate ($r = -.077, R^2 = .006, N = 702, p < .05$). However, the very low R^2 values indicate that proofreading time and scrolling amount accounted for little variance in predicting typo detection rate.

DISCUSSION

Proofreading Time

Line length has been shown to influence reading (not proofreading) speed. However, the results of research investigating the effects of line length have been inconclusive. Some authors have recommended long lines using measures such as reading speed and subjective preference (Dyson & Kipping, 1998; Ling & van Schaik, 2006). Some studies have found that shorter line length is preferable to longer (Shaikh & Chaparro, 2005), whereas others have found a preference for a medium line length, ranging from 45 to 65 characters per line (Davidov, 2002; Dyson & Haselgrove, 2001). With such mixed results, perhaps it is not so surprising that line length was not a significant factor on proofreading time here.

So far, not much research has focused on line number, but the result here is not consistent with the previous research for English (not proofreading), where it was found that reading speed

TABLE 5: Pearson Correlation Coefficients Among Proofreading Time, Detection Rate, and Amount of Scrolling (*N* = 702)

	Proofreading Time	Detection Rate	Scrolling Amount
Proofreading time	1		
Detection rate	.171***	1	
Scrolling amount	-.109**	-.077*	1

*Significant at the .05 level. **Significant at the .01 level. ***Significant at the .001 level.

increased if the line number was four or greater (Duchnick & Kolers, 1983) or seven or greater (Elkerton & Williges, 1984). Overall the results of these previous studies suggest an optimum line number of around five for window display and further increase of line number up to around 20 did not diminish or improve reading performance significantly. In related work, Neal and Darnell (1984) found that full page displays had little advantage over partial page displays, and Richardson, Dillon, and McKnight (1989) could not find any significant advantages with different line numbers. However, readers had a strong preference for showing more lines on screens. Here, eight lines resulted in the worst proofreading time. This rather unexpected result may be accounted for to some extent by the difference between reading and proofreading. Here, comparison proofreading was with top-down placement, which, unlike simply reading, requires comparison of live and dead copies, and this involves frequent eye movements between the two copies. With an increase in line number, the travel distance between the two copies increased; therefore, the time and difficulty of comparing the passages also increased. This might make it more difficult to accurately locate the start of the relevant next lines of the two copies, and might have some effect on eye fatigue and led to the finding of longer proofreading time with eight lines. However, the four lines did not produce the second longest proofreading times, despite the larger eye travel distance than for two lines. The faster proofreading times for four lines than for two lines could be attributed to the effect of scrolling during proofreading. For two-line proofreading, the screen size was by far the smallest, and it was therefore more prone to overscrolling, leading to the need for extra upward scrolling, accounting for more than half

of the total amount (57.79%) of upward scrolling. The extra upward scrolling was needed to relocate the text line before further proofreading, hence making proofreading times slightly longer than those for four lines, even though the travel distance for two lines was smaller than for four lines.

For the effect of line spacing on time, previous work in English was for reading and not proofreading. Muter and Maurutto (1991) recommended double spacing rather than single spacing for reading on computer screens. Chan and Lee (2005a) found that double spacing resulted in a slight increase in reading rate when comparing with single spacing. If line spacing is too close, the result is crowding that may affect the subjective experience of readers (Cline, Hofstetter, & Griffin, 1997). Crowding has been shown to be unfavorable for reading and that reading speed might increase as line spacing increased (Ling & van Schaik, 2007; Sheedy, Subbaram, Zimmerman, & Hayes, 2005). Chung (2004) has suggested that for English reading, double spacing might not be necessary and 1.5 line spacing was enough for facilitating reading. The trend for longer proofreading time with increased line spacing was found in this experiment. This result was not in agreement with previous research, most of which suggested a shortening of reading time as line spacing increased (Kolers, Duchnick, & Feguson, 1981; Kruk & Muter, 1984). Again, this result may be explained by the different line of sight travel distances required for the different levels of line spacing in proofreading. When the line spacing increased, so did line of sight travel distance, making the process of comparison more difficult and time-consuming.

For proofreading time, the only significant interaction was for line number and line spacing.

There was a general trend for increasing time with increasing line spacing; however, there were two conditions, eight-line with double spacing and four-line with single spacing, which did not match the main trend. For two lines, it was the least crowded condition for proofreading; changing the line spacing for two lines would affect only the eye movement distance but not have much effect on the crowding of the passage (Ling & van Schaik, 2007; Ma, Chan, & Courtney, 2004). This expectation was confirmed by the steady increasing trend of proofreading time for two lines with increasing line spacing. However, for four and eight lines, more lines were displayed on the screen and the effect of crowding became obvious. It was expected that the effect of line of sight travel distance and crowding of a passage would be compensatory in that a decrease in crowding would cause an increase in travel distance and vice versa. Such a compensatory effect might help explain why the conditions of eight-line with double spacing and four-line with single spacing did not conform to the main trend shown by the two-line condition. The lower than expected proofreading times for eight-line with double spacing might be attributable to the fact that the negative effect of increase in travel distance was offset by the positive effect of decrease in passage crowding when line spacing was increased to double spacing. In the case of four-line with single spacing, although the line of sight travel distance for single spacing was the lowest among different line spacing conditions, the crowding effect was the highest in this condition, leading to the higher than expected proofreading times for this condition, seemingly as a result of the domination of the crowding effect over the influence of travel distance. Of interest, in contrast to eight lines, further increase in line spacing from 1.5 to double for four lines did not result in faster proofreading times. This finding might be explained by the fact that the crowding with four lines was not as severe as with eight lines, and the further increase in line spacing and thus travel distance would offset the benefits from reduction in crowding, resulting in an increase in proofreading times.

Typo Detection Rate

For the significant line number factor, the highest detection rate was for four lines fol-

lowed by two lines, with eight lines being worst (Table 3). Elkerton and Williges (1984) suggested that the reading accuracy increased with passage of seven lines or more. The result here, where detection rate increased from two to four lines but decreased for eight lines, does not agree with their finding. The explanation is the same as for proofreading time. Participants here were proofreading but not reading, and proofreading requires line of sight movements between live and dead copies. The eight-line condition required longer line of sight travel distance than the four- or two-line conditions, making comparisons more difficult. It should follow then that two lines should have the highest detection rate, but this was not the case; the detection rate was lower than for four lines. However, two-line passages resulted in the greatest amount of scrolling. Some previous research suggests that more scrolling movement may result in distraction to readers, which could reduce reading accuracy (O'Hara & Snellen, 1997). Also, the correlation results here showed that more scrolling resulted in a lower detection rate, maybe because there was more distraction for two lines than for four lines.

For line spacing, there was a trend of increasing detection rate with increasing line spacing. This result is consistent with previous work on reading that increased line spacing resulted in increased reading accuracy (Ling & van Schaik, 2007). This effect may be because single spacing is relatively crowded and there was probably greater interference from surrounding characters, making it harder to locate text accurately and thereby increasing the difficulty of seeing embedded typos. The difference in detection rates between 1.5 and double spacing detection was not large, and this result is consistent with previous research showing that the trend of improving reading performance would slow down after line spacing increased to 1.5 (Chung, 2004). For detection rate, there was an interaction between line number and line spacing, which might be accounted for by the relatively stable detection performance for the text crowded eight-line condition with different line spacings (Figure 5).

Amount of Scrolling

For amount of scrolling, ANOVA showed that line length and line number had significant

effects but line spacing did not, and there were no interaction effects. Line length had no effect on proofreading time and detection rate, but it did affect scrolling, such that the average amount of scrolling decreased with increasing line length (Table 4). This result agrees with previous work on reading that longer line length resulted in less scrolling (Dyson & Kipping, 1998). This outcome seems reasonable and predictable because longer length results in fewer lines and line length obviously has a direct relationship with the screen size. For line number, scrolling amount decreased with increasing line numbers, most likely because line number directly affected the amount of text presented on screen. Only 7.76% of scrolling was upward, probably due in part to overscrolling. This result implies that there was not a lot of rereading previous passages. The overwhelming use of downward movement explains the correlations between scrolling amount and proofreading time (elaborated later). Use of the computer mouse scroll wheel and the screen scroll bar was greatly preferred (49.56% and 41.70%, respectively) to use of the keyboard (8.74%).

Further Remarks and Limitations

The main purpose of this study was to investigate the effects and interactions of line length, line number, and line spacing on screen-based Chinese proofreading performance. Other factors such as number of columns, display size, and resolution were not considered or examined. In a recent study by Ganayim and Ibrahim (2012), the column setting and line length showed a significant interaction effect on the overall comprehension scores. The mean comprehension score for one column with long line length was the highest, followed by that for two columns with medium line length, and then three columns with short line length. Future studies may explore the interaction of column setting and line length on proofreading performance to examine the possibility of further optimizing design parameters for on-screen Chinese proofreading.

Lin, Wu, and Cheng (2013) discovered that screen size and character size produced effects on search time and accuracy rate of a Chinese reading task. On the 6-in. and 9.7-in. screens, 12 point

took the shortest search time whereas the 8.1-in. screen was suitable for both 12 point and 14 point. With regard to accuracy rate, the 6-in. screen was suitable for 8 point whereas 9.7-in. screen was suitable for both 12 point and 14 point. Huang et al. (2009) also found significant differences in participants' reading performance and preference for different display resolutions and font sizes. Thus, the findings here might not be the same for smaller/bigger display, lower/higher resolution, and smaller/larger font size. Further studies may be conducted to examine the influence of display size and resolution on proofreading performance to further optimize design parameters for on-screen Chinese proofreading. No eye movement data were measured or analyzed in this study. The characteristics of eye movement measures in the two copies should be useful for understanding and gaining insight of the comparative proofreading strategies employed for the task by the participants. More information would then be provided for revealing if a bottom-up cognition or a top-down approach was more commonly practiced during proofreading in different display conditions.

A past study on reading Chinese fonts from mobile devices showed that subjective evaluation was also a sensitive measure as the objective measures (Huang et al., 2009). To optimize design parameters for on-screen Chinese proofreading, subjective evaluation may need to be done for understanding the preference of participants in addition to the measurement of objective measures on speed and accuracy in future.

Most of the participants in this study were males, which might have influenced the generalization of results in proofreading tasks. In the study of Liu and Huang (2008), decreased sustained attention in the online environment was found amongst the female readers, and male readers read things more selectively and had a greater degree of satisfaction with online reading. Taki and Soleimani (2012) also revealed that males and females differed in online reading strategies. Thus, for obtaining more representative findings, the numbers of male and female participants in the study should have been more balanced. Besides, the method used here for selecting the testing passages with similar difficulty was rather subjective. A more objective approach such as using a

perceived difficulty scale or a readability scale (Bell, Miller, McCallum, Hopkins, & Hilton-Prillhart, 2012; Heydari & Riazi, 2012) could have been better used to equate the difficulty of the passages. Also, it is noted that the method of reversing the sequence of the nine display conditions for testing may not be a good way of controlling potential sequencing effects. A better method of randomizing or counterbalancing the testing conditions should have been adopted instead.

With the advancement of computer technology and prevalence of its application in human work, there is an increasing trend to use different varieties of technology such as a dual-monitor configuration, a large display, a touch screen, and/or a tablet in a work environment. The findings of the three display factors, namely line length, line number, and line spacing, on proofreading tasks here might be changed under these situations. For example, the increase of display size may result in larger font size and larger window size, but inevitably increases line of sight travel distances between live and dead copies, which may not result in better proofreading performance. Future studies on proofreading tasks should thus consider the impact of technology on study design and the levels to be tested.

In terms of cultural differences and schemata, Singhai (1998) indicated that reading in the first language shares some basic elements with reading in a second or foreign language. Thus, reading in the first language can be similar to and can be very different from reading in the second language. The results of research investigating the effects of first and second language have not been conclusive. The readability of information designed for Chinese and used by readers of Chinese as a second or foreign language thus needs to be investigated. Regarding the writing system, it needs to be emphasized that not all the languages are written from left to right. Arabic, like most Semitic languages, is written from right to left. Readers may be interested in studies about the readability of documents or web pages designed for Latin alphabets or Chinese characters but used by people reading from right to left in other countries.

CONCLUSION

As a result of this study it is recommended that, for balancing time and detection rate

and improving performance for on-screen Chinese proofreading, a display setting of medium line length (36 characters per line) with four lines and 1.5 line spacing should be used. All these are new findings, and they can provide important information and recommendations for display factors and screen design to improve proofreading time and accuracy.

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KEY POINTS

- Proofreading time and typo detection rate increased with increasing line spacing, probably as a result of an increase in travel distance between the two copies.
- Two lines and single spacing resulted in the fastest proofreading time, whereas four lines and double spacing resulted in the highest typo detection rate.
- Scrolling amount decreased with increasing line number and line length but remained unchanged by varying line spacing.
- A display setting of medium line length (36 characters per line) with four lines and 1.5 line spacing is recommended for Chinese on-screen proofreading.

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